

Darin Tsui

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Education

Georgia Institute of Technology , Ph.D. in Electrical and Computer Engineering	Aug 2023 – May 2028
• Advisor: Amirali Aghazadeh	
University of California San Diego , B.S. in Bioengineering	Sep 2019 – Jun 2023

Selected Publications

Darin Tsui, Kunal Talreja, Daniel Saeedi, and Amirali Aghazadeh. Protein circuit tracing via cross-layer transcoders. *arXiv preprint arXiv:2602.12026*, 2026.

Zain Shabeeb*, Daniel Saeedi*, **Darin Tsui***, Vida Jamali, and Amirali Aghazadeh. cryoSENSE: Compressive sensing enables high-throughput microscopy with sparse and generative priors on the protein cryo-EM image manifold. *arXiv preprint arXiv:2511.12931*, 2025.

*Equal contribution.

Darin Tsui, Aryan Musharaf, Justin S. Kang, Yigit E. Erginbas, and Amirali Aghazadeh. SHAP zero explains biological sequence models with near-zero marginal cost for future queries. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

Honors & Awards

Steve W. Chaddick Fellowship , Georgia Tech	2025
• Fellowship awarded for outstanding contributions in research.	
NSF Graduate Research Fellowship	2024 – 2027
• Fellowship recognizing potential for significant research contributions (top 14% of applicants).	
Course Enhancements & Improvement (CREATION) Award , Georgia Tech	2023
• Award received for developing the course ECE 8803— <i>Generative and Geometric Deep Learning</i> .	
Graduate Student Conference Scholarship Award , Conference on Information Storage and Processing Systems (ISPS)	2023
• Award received for outstanding paper submission and oral presentation.	
President's Fellowship , Georgia Tech	2023
• Fellowship awarded to top 10% of Ph.D. applicants based on their research and scholarship.	
Anushka Michailova Memorial Bioengineering Award , UC San Diego	2023
• Award received for best capstone project out of 100 teams. Submissions were evaluated on technical depth, innovation, and research dissemination.	

Research Experience

AI ML and Information Research Group , Georgia Tech – Atlanta, GA	Jun 2023 – Present
• Advisor: Amirali Aghazadeh.	
• Leveraged <i>mechanistic interpretability</i> frameworks, such as cross-layer transcoders and sparse autoencoders, for circuit discovery, protein design, and extracting biological mechanisms from protein language models.	
• Extracted higher-order interactions in models 1000-fold quicker than state-of-the-art methods by developing algorithms at the intersection of <i>signal processing</i> , <i>coding theory</i> , and <i>combinatorics</i> .	
• Developed principled <i>generative models</i> for high-throughput cryo-electron microscopy (cryo-EM), autoregressive protein generation, and variant effect prediction.	
Integrated Systems Neuroengineering Lab , UC San Diego – San Diego, CA	Jun 2022 – Jun 2023
• Advisor: Gert Cauwenberghs. Mentored by Ph.D. candidates Akshay Paul, Tyler Bodily, and Min Lee.	

- Designed and developed COVID-19 diagnostic test using a graphene field-effect-transistor (GFET).
- Integrated feature extraction and machine learning pipeline for current-voltage signals.
- Achieved accuracies of 98.5% when detecting COVID-19 proteins, improving classification by 30.1%. Presented and published work at the *2023 IEEE Engineering and Medicine and Biology Conference (EMBC)*.

Talke Biomedical Device Lab, UC San Diego – San Diego, CA

Dec 2021 – Aug 2023

- Advisor: Frank E. Talke
- Designed low-cost vision system for minimally invasive surgery using OpenCV with fiducial markers.
- Integrated the tracking of ArUco markers on surgical platforms using Python and MATLAB.
- Implemented Kalman filtering to achieve sub-millimeter error in design-validation testing. Presented and published work in the journal *Microsystem Technologies* and the conferences *ISPS 2023* and *EMBC 2023*.

Swartz Center for Computational Neuroscience, UC San Diego – San Diego, CA

Mar 2021 – Jun 2023

- Advisor: Tzyy-Ping Jung. Mentored by Ph.D. candidate Chiyuan Chang.
- Designed electroencephalography-based (EEG) classification experiments. Conducted machine learning analysis of classroom stress levels.
- Performed support vector machine (SVM) and linear discriminant analysis (LDA) and Mann Whitney U statistical analysis using MATLAB and Python.
- Improved classification of stress vs. non-stress groups by 42.1% in correlation with physiological features.

Paesani Research Group, UC San Diego – San Diego, CA

Jan 2020 – Sep 2021

- Advisor: Francesco Paesani. Mentored by postdoctoral scholar Vinícius Cruzeiro.
- Designed and ran AMBER 20 molecular dynamics simulations in Bash and Python on supercomputing clusters to model amino acid and water molecule behaviors.
- Validated interactions using radial distribution function (RDF) fits.

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Darin Tsui, Kunal Talreja, Daniel Saeedi, and Amirali Aghazadeh. Protein circuit tracing via cross-layer transcoders. *arXiv preprint arXiv:2602.12026*, 2026.

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Thomas A. Walton, **Darin Tsui**, Aryan Musharaf, and Amirali Aghazadeh. SpecMER: Fast protein generation with K-mer guided speculative decoding. In *NeurIPS (Spotlight)*, 2025.

Darin Tsui, Kunal Talreja, and Amirali Aghazadeh. Sparse autoencoders for low- N protein function prediction and design. In *NeurIPS AI4Science Workshop*, 2025.

Thomas A. Walton, **Darin Tsui**, Lauren Fogel, Dustin J. E. Huard, Rafael S. Chagas, Raquel L. Lieberman, and Amirali Aghazadeh. GOLF: A generative AI framework for pathogenicity prediction of myocilin OLF variants. In *Machine Learning in Computational Biology*, 2025.

Darin Tsui, Kunal Talreja, and Amirali Aghazadeh. Efficient algorithm for sparse Fourier transform of generalized q -ary functions. In *IEEE Information Theory Workshop (ITW)*, 2025.

Darin Tsui and Amirali Aghazadeh. On recovering higher-order interactions from protein language models. In *International Conference on Learning Representations (ICLR) Generative and Experimental Perspectives for Biomolecular Design (GEM) Workshop*, 2024.

Teaching Experience

Teaching Assistant, Georgia Tech – Atlanta, GA

Aug 2023 – Dec 2023

- Advisor: Jeffery Hurley
- Graduate teaching assistant for ECE 4122/6122—*Advanced Programming Techniques for Engineering Applications*—with an enrollment of 300 graduate and undergraduate students. The course teaches undergraduate and graduate students techniques for distributed and parallel computing to run computational heavy tasks.
- Taught and assisted students in developing C++ code for lab assignments. Held office hours for 10 hours a week and mentored students on their final projects.

Instructional Assistant, UC San Diego – San Diego, CA

Dec 2020 – Mar 2021

- Advisor: Robert Sah
- Oversaw two lab sections of 30 students each for BENG 1—*Introduction to Bioengineering*.
- Taught lab sections covering topics ranging from biomechanics, noninvasive cardiovascular sensing, and spine segmentation. Held office hours and lab sections for 10 hours a week.

Industry Experience

Data Engineering Intern, Surgalign – San Diego, CA

Jan 2023 – Aug 2023

- Manager: Weidong Luo
- Interned at Surgalign, a small medical device company focused on developing AI solutions in spinal surgery (now acquired by Xtant Medical and Augmetics).
- Designed an internal MRI image processing application using Python that reduced preprocessing time.
- Assisted in developing ground truth dataset of imaging data and validated deep learning models against surgeon-generated data using Sørensen-Dice coefficient statistical testing.

Instruments R&D Intern, Johnson & Johnson – Redwood City, CA

Jun 2022 – Aug 2022

- Manager: Timothy Reppert
- Interned at Auris Health, Johnson & Johnson's flexible robotics division for minimally invasive surgery of urology. Supported design validation clinical trials through preparation and sterilization of instruments.
- Performed root cause analysis on the failure of instrument devices. Designed near-infrared camera system for calibration in endoscopes.
- Supported systems investigations towards optical calibration stations and instrumentation platforms.

Presentations

Darin Tsui. SHAP zero explains biological sequence models with near-zero marginal cost for future queries, *presented at the ML Student Seminar*, Atlanta, GA, 2025.

Darin Tsui. Explainability and mechanistic interpretability in AI, *presented as a guest lecturer for ECE 8803—Generative and Geometric Deep Learning—at Georgia Tech*, Atlanta, GA, 2025.

Darin Tsui. Generative and geometric deep learning in computational biology, *presented as a guest lecturer for ECE 8803—Generative and Geometric Deep Learning—at Georgia Tech*, Atlanta, GA, 2023.

Darin Tsui. Surgical site localization with non-invasive skin markers for pain management, *presented at the Center for Memory and Recording Research (CMRR) Research Review*, San Diego, CA, 2023.

Darin Tsui. Bioelectronic COVID-19 detection via graphene-field-effect-transistor (GFET), *presented at Bioengineering Day at UC San Diego*, San Diego, CA, 2023.

Darin Tsui. Design of a surgical navigation system via positional tracking of fiducial markers, *presented at the Conference on Micromechatronics for Information and Precision Equipment*, San Diego, CA, 2022.

Service

Reviewer: 2026 ICML, CVPR, ECCV, ICLR GEM; 2025 NeurIPS AI4Science; 2024 ICLR GEM.

Bioengineering Day Chair, Georgia Tech – Atlanta, GA Jan 2025 – Present

- Co-organized Bioengineering Day, a symposium highlighting interdisciplinary research with over 100 attendees.
- Coordinated logistics, speaker outreach, and scheduling for faculty and alumni talks, award presentations, and a rapid-fire thesis competition.

Capstone Project Advisor, UC San Diego – San Diego, CA Jul 2023 – Jun 2024

- Capstone project advisor for a bioengineering capstone project with Dr. Frank E. Talke.
- Mentored two undergraduates toward the design and development of a surgical navigation system with 3D tracking and augmented reality.
- Capstone project team won the *Teaching Assistant's Choice Award* at Bioengineering Day.

Founding Member, IEEE EMBS at UC San Diego – San Diego, CA Apr 2023 – Jul 2023

- Co-founded the IEEE Engineering in Medicine and Biology Society (EMBS) at UC San Diego with Dr. Gert Cauwenberghs and other Ph.D. students to increase awareness of bioengineering topics and research.
- Recruited 5 graduate students to mentor undergraduate students interested in getting involved in research.

President, IEEE at UC San Diego – San Diego, CA May 2022 – May 2023

- Oversaw operations for a 300+ student body. Partnered with the San Diego Supercomputing Center (SDSC) to co-found IEEE's Supercomputing Team.
- Hosted lab recruiting talks and technical seminars targeted towards first and second-years looking to get into research. Topics included research in artificial intelligence, robotics, and wearable sensors.
- Awarded the Outstanding IEEE Large Student Branch award for maintaining strong project activities and company relations during tenure.

Outreach Chair, IEEE at UC San Diego – San Diego, CA May 2021 – May 2022

- Taught 60 elementary and middle school students programming basics at STEM Fair.
- Hosted a robotics workshop in collaboration with Robolink, a San Diego robotics company, where we invited 50 high school students from low-income communities to work with Robolink representatives on coding and designing their own aerial drones.

Project Team Lead, BMES at UC San Diego – San Diego, CA Oct 2021 – Jun 2022

- Led team of 20 undergraduates in designing a telesurgical prosthetic hand at UC San Diego's Biomedical Engineering Society (BMES). Hand designs were formulated and constructed based on biomechanics principles.
- Held workshops in CAD, Arduino, and circuit design. Implemented robotic motion in Arduino and interfaced linear actuators.

Committee Member, BMES at UC San Diego – San Diego, CA Sep 2019 – Jun 2023

- Committee member for BMES's Bioengineering Day committee, a day-long event celebrating all facets of bioengineering research and development, and Lab Expo committee, an undergraduate research symposium aiming to get students involved in research and academia.
- Recruited graduate students and professors to present research. Organized logistics and advertised events.

Mentoring

William Deinzer, CS Undergraduate, Georgia Institute of Technology	2026
Rishi Manimaran, CS Undergraduate, Georgia Institute of Technology	2025 – 2026
Kunal Talreja, ECE Undergraduate, Georgia Institute of Technology	2024 – 2026
Aryan Musharaf, CS Undergraduate, Georgia Institute of Technology	2024 – 2025
Matthew Tam, Bioengineering Undergraduate, UC San Diego	2023 – 2024
Kirsten Ramos, Bioengineering Undergraduate, UC San Diego	2023 – 2024
Eric Lee, Bioengineering Undergraduate, UC San Diego	2023 – 2024

Capalina Melentyev, Bioengineering Undergraduate, UC San Diego
Rohan Kumar, Nanoengineering Undergraduate, UC San Diego

2022 – 2024

2022 – 2023